GURMILAN SINGH

Full-Stack Web Developer | Software Engineer | React · JavaScript · Node.js · SQL · C#/.NET

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PROFESSIONAL SUMMARY

Final-year Computer Engineering student and software developer with 3+ years of professional experience building and maintaining production full-stack applications. Primary developer of an internal business platform, with end-to-end responsibility spanning frontend and backend development, relational database design, authentication, deployment, maintenance, and user support. Experienced with C#/.NET, React, Node.js, SQL, automation, infrastructure support, and AI/API integrations.

PROFESSIONAL EXPERIENCE

Software Developer — C#/.NET & Full-Stack Web

Phema S.r.l. · Modena, Italy · 09/2023 – Present

React · JavaScript · Material UI · Node.js · Express · SQL Server · Jest · C#/.NET · systems administration

- Primary developer of an internal full-stack business platform supporting shared operational workflows, role-based access and centralized relational data.
- Own development across requirements analysis, frontend/backend implementation, database design, authentication, testing, releases, maintenance and user support.
- Helped replace disconnected legacy tools and manual processes with a centralized application covering operational records, scheduling, project coordination, reporting and administration.
- Contribute to the company's commercial C#/.NET software, including a client-server licence-management component, while supporting the application's server and virtualized environment.

SELECTED PROJECTS

MaintOps — Maintenance Operations System

React · Fastify · MySQL · REST · RBAC · Vitest

- Designed and built a full-stack maintenance system covering request triage, technician workflows, assignments, SLA tracking, reporting and audited administration.
- Implemented server-side authorization, transactional state transitions, versioned SLA policies, background workers and protected attachment delivery.
- Added optional task-specific AI assistance behind bounded context, structured-output validation and human approval, while keeping all core workflows independent from the AI provider.

Opssemble — AI-Assisted Event Operations Platform

React 18 · Vite · Node.js · Fastify · AI APIs · JWT · REST API · Nodemailer · MySQL · Google OAuth 2.0

- Built a full-stack event operations platform for multi-user planning, area coordination, task tracking, exception management, readiness scoring, reporting, and scoped QR staff access.
- Designed a modular Fastify/MySQL REST API using a routes/controller/service/repository/schema structure across authentication, events, members, areas, tasks, exceptions, readiness, reports, activity logs, QR access, AI, templates, settings, and admin modules.
- Implemented secure authentication with email verification, password reset, Google OAuth 2.0, JWT-backed HttpOnly cookies, bcrypt hashing, generic auth errors, failed-login throttling, and account lockouts.
- Built QR-code area access using revocable SHA-256-hashed tokens, allowing on-site staff to access scoped event areas while preserving backend permission checks.
- Integrated a pluggable AI assistant using hosted Groq, supporting task/exception drafting, event scaffolding, gap checking, readiness explanations, blocker summaries, event Q&A, and retrospective drafting.
- Added production-style engineering features including self-healing MySQL migrations, activity audit logs, readiness scoring, CSV/print reports, transactional emails, upload validation by magic bytes, Helmet security headers, CORS policies, rate limiting, and parameterized SQL queries.

Gurmat Saanj — Full-Stack Web App with Remote Control System

React · JavaScript · Vite · Node.js · Express · BaniDB · Web Speech API

- Built a deployed application that follows live Punjabi recitation, matches recognized phrases against Gurbani lines and synchronizes selected content across projector and viewer interfaces.
- Implemented QR/session-based mobile control and real-time synchronization while supporting device-local presentation preferences.

- Hardened API boundaries with validation, rate limiting, CORS and security middleware.

Alfaaz — Private Multilingual Writing System

React · Express · MySQL · Zod · Vitest · Playwright

- Building a multilingual writing environment for English, Hindi and Punjabi with autosave, optimistic conflict handling, revision history, collections and local transliteration assistance.
- Designed privacy boundaries so unfinished transliteration, local music, focus-session state and generated exports can remain client-side when server persistence is unnecessary.
- Implemented authenticated user-scoped storage, protected voice-note delivery and automated client, server and end-to-end testing.

ADDITIONAL PROJECTS

Azure VM Provisioning Automation

Bash · Azure CLI · Linux (Ubuntu) · SSH · University project

- Wrote **Bash (.sh) scripts to automate provisioning of Ubuntu virtual machines on Microsoft Azure** over SSH, replacing a manual setup process with a repeatable, scripted workflow.

SIRD Epidemic Simulator — Java Desktop Application

Java 17 · JavaFX · Maven · FXML · Concurrency

- Agent-based epidemic simulation with a background engine, real-time S/I/R/D visualization, configurable interventions and CSV export.

Esamarea — Multi-Exam Study Planning System

TypeScript · React 19 · Express 5 · Prisma · MySQL · Vitest · Playwright

- Full-stack study planner that distributes shared weekly availability across multiple exams, generates deterministic plans and records server-timed study sessions, with UI and end-to-end testing.

TECHNICAL SKILLS

Languages: C#, TypeScript, JavaScript, Java, Python, SQL, C, Bash
Backend: ASP.NET Core, .NET, Entity Framework Core, Node.js, Express, Fastify, REST APIs, JWT, RBAC
Frontend: React, Vite, HTML5, CSS3, Material UI, TanStack Query
Databases: SQL Server, MySQL, relational database design, migrations
Testing: Vitest, Playwright, Supertest, Jest, Node.js Test Runner, e2e
Systems & Delivery: Linux, Docker, Azure, Azure CLI, VMware ESXi, Git, GitHub Actions, SSH, Postman
AI & Integrations: LLM API integration, Ollama, Groq, Google OAuth 2.0, Web Speech API

EDUCATION

BSc in Software Engineering - Laurea in Ingegneria Informatica
University of Modena and Reggio Emilia (UNIMORE) · 2023 – Present · Expected December 2026

Final year. Core areas: programming, algorithms, data structures, databases, networks, operating systems, software engineering, web technologies, and machine learning / AI.

Diploma in Computer Science and Telecommunications
ITIS Enrico Fermi, Modena · 2018 – 2023

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Cisco Networking Academy — Introduction to Cybersecurity, 2026
- IBM SkillsBuild — Getting Started with Cybersecurity, 2026
- IBM SkillsBuild — Getting Started with Data, 2026
- Udemy — The Web Developer Bootcamp 2024, Colt Steele, 74 hrs
- Self-directed online coursework in C#, C, Java, Python, and React

LANGUAGES

- Italian: Native · Punjabi: Native · English: C1 Advanced · Hindi: Advanced

Additional information: Italian driving licence, category B

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